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ANNEXURE A

ERROR ANALYSIS TASK

Given Confusion Matrix

Actual \ Predicted	Cat	Dog	Bird	Rabbit
Cat	42	5	2	1
Dog	6	38	3	3
Bird	1	4	43	2
Rabbit	2	3	5	40

Total images = **200**

1. Calculate the Overall Accuracy of the Model

The overall accuracy measures how many images were classified correctly out of the total number of test images.

Formula:

Accuracy=Number of Correct PredictionsTotal Number of Predictions×100\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100Accuracy=Total Number of PredictionsNumber of Correct Predictions×100

The correct predictions are the diagonal elements of the confusion matrix:

- Cat correctly classified = **42**
- Dog correctly classified = **38**
- Bird correctly classified = **43**
- Rabbit correctly classified = **40**

Total correct predictions:

42+38+43+40=16342 + 38 + 43 + 40 = 16342+38+43+40=163

Total test images:

200200200

Therefore,Accuracy=163200×100=81.5%\text{Accuracy} = \frac{163}{200} \times 100 = 81.5\%Accuracy=200163×100=81.5%

Answer:

The CNN model correctly classified **163 out of 200 images**, resulting in an **overall accuracy of 81.5%**. This means the model correctly predicts approximately **82 out of every 100 images**, indicating good performance but leaving room for improvement.

2. Identify Which Class Has the Highest Number of Misclassifications

Misclassification means the model predicted the wrong class.

Misclassifications for each class are calculated as:

Cat

Total Cat images = 50

Correct = 42

Misclassified = 50 − 42 = **8**

Dog

Total Dog images = 50

Correct = 38

Misclassified = 50 − 38 = **12**

Bird

Total Bird images = 50

Correct = 43

Misclassified = 50 − 43 = **7**

Rabbit

Total Rabbit images = 50

Correct = 40

Misclassified = 50 − 40 = **10**

Comparison

Class	Misclassifications
Cat	8
Dog	**12**
Bird	7
Rabbit	10

Answer:

The **Dog** class has the highest number of misclassifications (**12**). This indicates that the model finds it more difficult to correctly recognize dog images than the other classes.

3. List the Two Most Common Classification Errors

The largest off-diagonal values in the confusion matrix represent the most frequent mistakes.

From the matrix:

- Dog predicted as Cat = **6**
- Cat predicted as Dog = **5**
- Rabbit predicted as Bird = **5**

Most Common Errors

1. **Dog → Cat (6 images)**
2. **Cat → Dog (5 images)**

4. Suggest Three Possible Reasons Why These Errors Occurred

1. Similar Visual Features

Cats and dogs have similar appearances, including fur, four legs, tails, and facial structures. Because of these similarities, the CNN may struggle to distinguish between them, especially if images are taken from different angles or lighting conditions.

2. Insufficient or Imbalanced Training Data

If some classes contain fewer training images or lack diversity, the model cannot learn all possible variations. For example, if there are only a few rabbit images or very similar dog images, the CNN may fail to generalize well.

3. Poor Image Quality and Background Variations

Images with low resolution, blur, shadows, poor lighting, or complex backgrounds make feature extraction more difficult. Background objects can also distract the CNN and reduce classification accuracy.

5. Recommend Four Techniques to Improve Model Performance

1. Increase the Training Dataset

Collect more images for all four classes under different conditions such as lighting, camera angles, poses, and backgrounds. More data helps the CNN learn richer features and improves generalization.

2. Apply Data Augmentation

Artificially increase the dataset by applying transformations such as:

- Rotation
- Flipping
- Zooming
- Cropping
- Brightness adjustment
- Translation

This helps the model become more robust to real-world image variations.

3. Use Transfer Learning

Instead of training from scratch, use pre-trained CNN models such as:

- VGG16
- ResNet50
- MobileNet
- EfficientNet

These models have already learned powerful image features from millions of images and usually achieve higher accuracy with less training data.

4. Tune Hyperparameters

Optimize parameters such as:

- Learning rate
- Batch size
- Number of epochs
- Optimizer (Adam, SGD)
- Dropout rate

Proper tuning improves learning efficiency and reduces errors.

6. Explain Whether Collecting More Training Data or Increasing the Number of CNN Layers Would Be More Beneficial

Collecting **more training data** would be more beneficial than simply increasing the number of CNN layers.

Explanation

The confusion matrix shows that the model already recognizes many images correctly but still confuses similar classes like cats and dogs. This suggests that the main limitation is not necessarily the network architecture but the diversity of the training data.

Adding more images with different poses, lighting conditions, backgrounds, and viewpoints helps the CNN learn more representative features and improves its ability to generalize to unseen images.

Increasing the number of CNN layers makes the model more complex. If sufficient training data is not available, the deeper model may memorize the training data instead of learning meaningful patterns, leading to overfitting and poorer test performance.

Conclusion

Although deeper CNNs can improve performance, collecting more diverse training data is the better choice in this situation because it directly addresses the model's classification errors and improves generalization.

7. If the Model Achieves 99% Training Accuracy but Only 82% Testing Accuracy, What Problem Does It Suffer From? Explain How You Would Address It.

The model suffers from overfitting**.**

Explanation

Overfitting occurs when the CNN learns the training images extremely well, including noise and unnecessary details, instead of learning general patterns that apply to new images.

As a result:

- Training accuracy becomes very high (**99%**).
- Testing accuracy remains much lower (**82%**).
- The model performs well on known data but poorly on unseen data.

Methods to Address Overfitting

1. Data Augmentation

Generate additional training images by rotating, flipping, zooming, and cropping existing images.

2. Dropout

Randomly deactivate some neurons during training to prevent the network from relying too heavily on specific features.

3. Regularization (L2 Weight Decay)

Penalize very large weights so that the model learns simpler and more general patterns.

4. Early Stopping

Stop training when the validation accuracy stops improving, preventing the model from memorizing the training data.

5. Collect More Training Data

A larger and more diverse dataset helps the CNN learn better general features and reduces overfitting.

6. Reduce Model Complexity

If the CNN is unnecessarily deep, reducing the number of layers or parameters can improve generalization.

Conclusion

The difference between **99% training accuracy** and **82% testing accuracy** clearly indicates **overfitting**. Applying data augmentation, dropout, regularization, early stopping, increasing training data, and simplifying the model can significantly improve testing performance and make the CNN more reliable on unseen images.